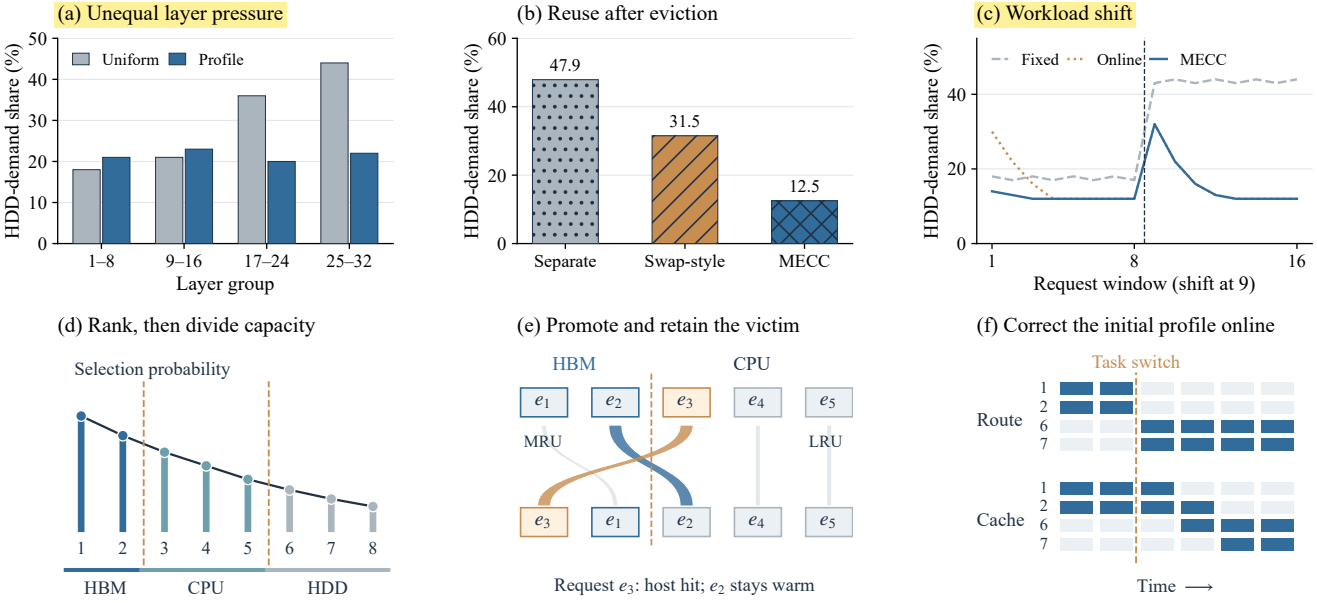

(a) Unequal layer pressure
HDD-demand share (%)
Uniform
Profile
1–8
9–16
17–24
25–32
Layer group
(b) Reuse after eviction
HDD-demand share (%)
47.9
31.5
12.5
Separate
Swap-style
MECC
(c) Workload shift
HDD-demand share (%)
Fixed
Online
MECC
Request window (shift at 9)
(d) Rank, then divide capacity
Selection probability
HBM
CPU
HDD
(e) Promote and retain the victim
HBM
CPU
MRU
LRU
Request e_3: host hit; e_2 stays warm
(f) Correct the initial profile online
Task switch
Route
Cache
Time